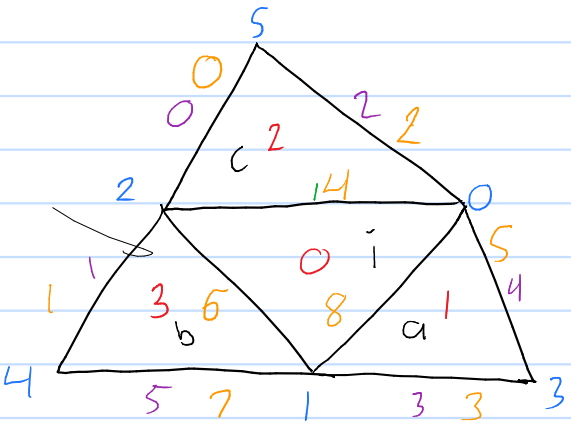


Cell
Edge
Boundary Edge
Vertex

Vertex	x	y
0	3.25	1.25
1	3	0.25
2	2	1
3	4.5	0.5
4	1.5	0
5	2.5	2.25

Edge lengths

0	1.3463
1	1.1180
2	1.25
3	1.5207
4	1.2748
5	1.4577
6	1.25
7	1.5207
8	1.0308

Cell Centroids

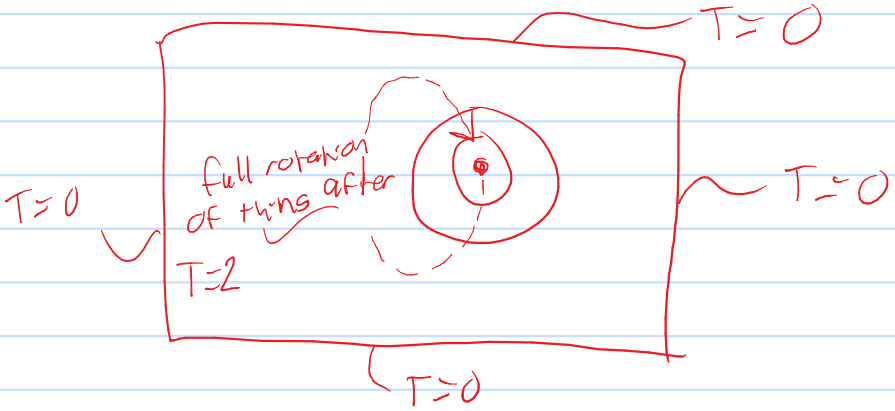
Cell	x	y
0	2.75	0.8333
1	3.5833	0.6667
2	2.5833	1.50
3	2.1667	0.4167

Edge Mid points

Edge	x	y
0	2.25	1.625
1	1.75	0.5
2	2.875	1.75
3	3.75	0.375
4	2.625	1.125
5	3.875	0.875
6	2.5	0.625
7	2.25	0.125
8	3.125	0.75

Apply T=0 for all boundaries

To check accuracy, run solution for time of period (2), & then compare the value to the initial solution the L2Norm of the difference of these two solutions will be the error



We also need the normal for each cell.

Remember, for edge i, with vertices A & B,

$$\hat{n}_i = \frac{1}{l_i}\begin{pmatrix} y_A - y_B \\ -x_A + x_B \end{pmatrix}$$

Based on this, & the lengths & coordinates calculated on the previous page, the following normal vectors can be calculated

Edge Normals

Edge	$\hat{n}_{x_i}$	$\hat{n}_{y_i}$	length	x_{A_i}	y_{A_i}	x_{B_i}	y_{B_i}
0	0.92847	-0.37139	1.3463	2.5	2.25	2	1
1	0.89443	-0.44721					
2	-0.8	-0.6					
3	-0.1644	0.98639					
4	-0.19612	0.98058					
5	-0.51450	-0.85749					
6	-0.6	-0.8					
7	-0.1644	0.98639					
8	-0.97011	0.24254					

Velocities at edge based on $u = \pi y$, $v = \pi x$

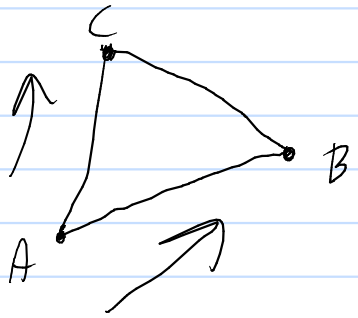
Edge	u	v	x	y
0	5.10511	-7.0686	2.25	1.625
1	1.157080	-5.49779		
2	5.49779	-9.03208		
3	1.17810	-11.78097		
4	3.53429	-8.24668		
5	2.74889	-12.17367		
6	1.96350	-7.85398		
7	0.39270	-7.06858		
8	2.35619	-9.81748		

T_0 at Cell centroid based on $T_0(x,y) = \exp(-5(x^2+(y-1)^2))$

Cell	T	x	y
0	3.2955e-17	2.75	0.8333
1	7.5234e-29	3.5833	0.6667
2	9.2378e-16	2.5833	1.5
3	1.1675e-16	2.1667	0.41667

Area

If ABC is



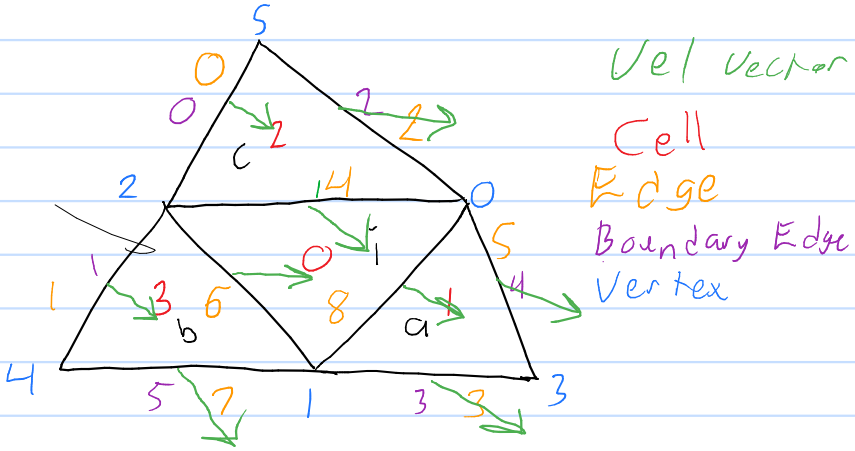
$\text{Area} = \frac{1}{2}(AB \times AC)$ positive

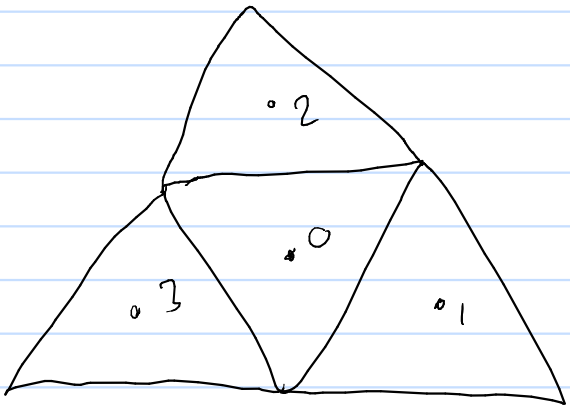
With triangle like this, if positive vertices are ordered CCW, if clockwise negative

$$AB \times AC = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ AB_x & AB_y & 0 \\ AE_x & AC_y & 0 \end{vmatrix} = 0\hat{i} - 0\hat{j} + (AB_x \cdot AC_y - AB_y AC_x)\hat{k}$$

Check cross products $(x_B - x_A)\cdot(y_C - y_A) - (y_B - y_A)(x_C - x_A)$

Cell	AB	C	x_A	y_A	x_B	y_B	x_C	y_C	AB_x	AB_y	AC_x	AC_y	$\lvert\frac{1}{2}AB \times AC\rvert$
0	0	1	2	3.25	1.25	3	0.25	2	1	-0.25	-1	-1.25	-0.25	0.5938
1	0	1	3	3.25	1.25	3	0.25	4.5	0.5					0.7188
2	0	2	5	3.25	1.25	2	1	2.5	2.25					0.7188
3	1	2	4	3	0.25	2	1	1.5	0					0.6875





for Cell 0 Gradient

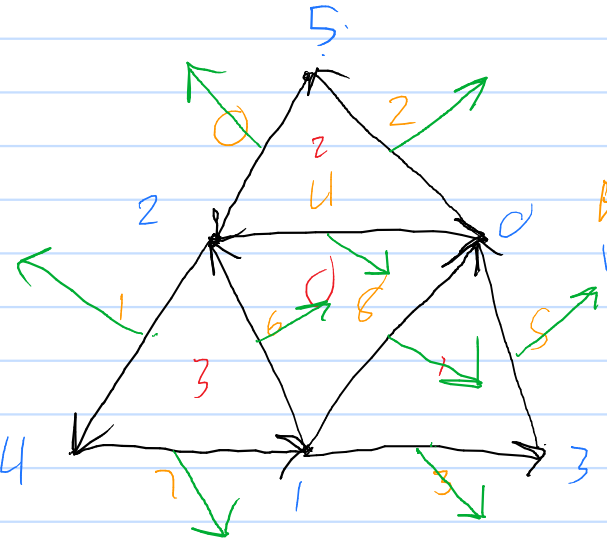
$$\Delta x_1 = x_1 - x_0 \qquad \Delta \bar{T}_1 = \bar{T}_1 - \bar{T}_0$$
$$\Delta x_2 = x_2 - x_0 \qquad \Delta \bar{T}_2 = \bar{T}_2 - \bar{T}_0$$
$$\Delta x_3 = x_3 - x_0 \qquad \Delta \bar{T}_3 = \bar{T}_3 - \bar{T}_0$$

$$\Delta y_1 = y_1 - y_0$$
$$\Delta y_2 = y_2 - y_0$$
$$\Delta y_3 = y_3 - y_0$$

$$\Delta x^2 = \Delta x_1^2 + \Delta x_2^2 + \Delta x_3^2$$
$$\Delta y^2 = \Delta y_1^2 + \Delta y_2^2 + \Delta y_3^2$$
$$\Delta x \Delta y = \Delta x_1 \Delta y_1 + \Delta x_2 \Delta y_2 + \Delta x_3 \Delta y_3$$

$$\Delta xT = \Delta x_1 \Delta T_1 + \Delta x_2 \Delta T_2 + \Delta x_3 \Delta T_3$$
$$\Delta yT = \Delta y_1 \Delta T_1 + \Delta y_2 \Delta T_2 + \Delta y_3 \Delta T_3$$

$$\begin{pmatrix} T_x \\ T_y \end{pmatrix} = \frac{1}{\Delta x^2 \Delta y^2 - (\Delta x \Delta y)^2} \begin{pmatrix} \Delta y^2 \Delta xT - \Delta xy \, \Delta yT \\ -\Delta xy \, \Delta xT + \Delta x^2 \Delta yT \end{pmatrix}$$



Normal
Cell
Edge
Vertex
Velocity

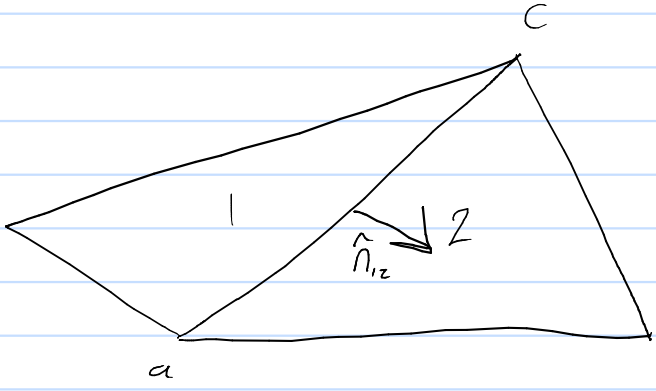
From this graph, we can see that as you walk from edge origin to destination, the normal points to our right. Therefore if the normal dot vel is positive, the upwind cell will be the left cell, & if it is negative, it will be the right cell.

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To for each cell in analysis

0	:	100
1	a	102
2	c	101
3	b	97

Edge AB in analytical is edge 8 in our mesh. The upwind cell in this case would be cell 0 (i), so we expect a solution for edge 8 of 100.95



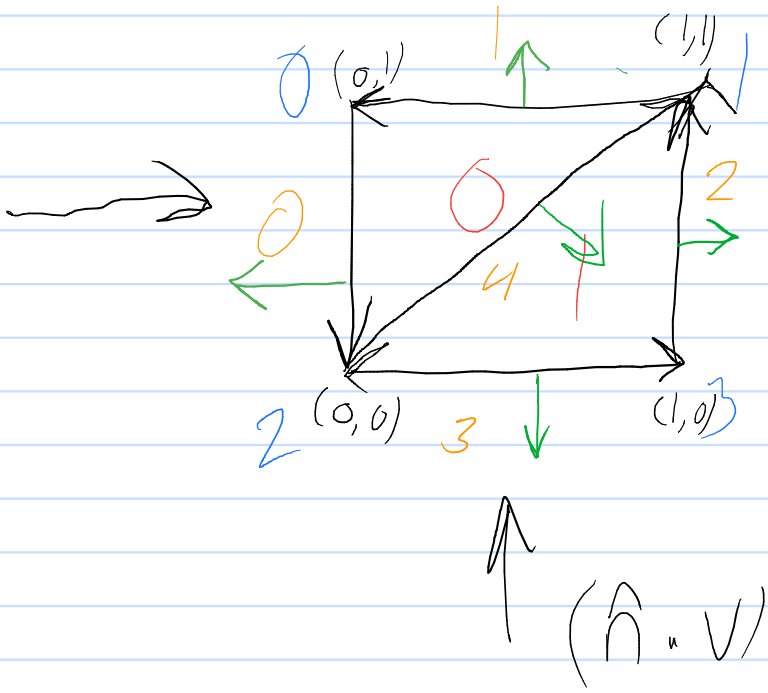
$$\hat{n} = \frac{1}{l_{ac}} \begin{pmatrix} y_c - y_a \\ -x_c + x_a \end{pmatrix}$$

Here a is origin, c is destination, & $\hat{n}$ points right as you go from a to c

$$\hat{n} = \frac{1}{l_{case}} \begin{pmatrix} y_{dest} - y_{origin} \\ -x_{dest} + x_{origin} \end{pmatrix}$$ & points into right cell

To calculate temperature at edge midpoint

$$T_i = \bar{T}_i + \frac{\partial T}{\partial x}\Big|_i (Edge_cent_x - Cell_Cent_X) + \frac{\partial T}{\partial y}\Big|_i (Edge_cent_y - cell_cent_y)$$



$$\vec{V} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \hat{n} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\hat{n} \cdot \vec{V} = 0 \cdot 0 + 1 \cdot 1 = 1$$

$$\vec{V} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{n} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\hat{n} \cdot \vec{V} = 1 \cdot 0 + 0 \cdot 1 = 0$$

$(\hat{n} \cdot V) / a_i \, l_e$

To check upwind cell calculation for each edge we can use the above geometry & impose a velocity of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ we know for $V_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ we will get a dot product of 1 for edges 0 & 3 & 0 for edges 1 & 4. The opposite is true for $V_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Finally we know with V_1 cell 0 will be upwind of edge 2, & cell 1 will be upwind for

file

2 5 4 4

0 1

1 1

0 0

1 0

0 -1 0 2

0 -1 1 0

1 -1 3 1

1 -1 2 3

0 1 2 1

$$f = \begin{pmatrix} uT \\ vT \end{pmatrix} \qquad u = 1,\ v = 0$$

$$T = 1 - x \Rightarrow f = \begin{pmatrix} uT \\ vT \end{pmatrix} = \begin{pmatrix} (1)(1-x) \\ (0)(1-x) \end{pmatrix} = \begin{pmatrix} 1-x \\ 0 \end{pmatrix}$$

$$\nabla \cdot f = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \end{bmatrix} \cdot \begin{pmatrix} 1-x \\ 0 \end{pmatrix} = -1 + 0 = -1$$

$$\overline{T}_i^{n+1} = \overline{T}_i^n$$

h	# cells	Δt
0.2	25	1.0
0.1	100	0.5
0.05	400	0.25